

How Do Lead Lag Relations in Cross Listings Move?

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This is a rather preliminary draft of a fairly comprehensive project. Please do not quote without explicit permission from the author. I gratefully acknowledge excellent programming assistance from Arko Bhattacharya.

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Abstract

We investigate the lead lag relations in portfolio returns, volatility, and spread. We test two comprehensive hypotheses on multi lag spillovers in returns, volatility, and spread using two sets of portfolios. The first set of portfolios are matched ADRs and their corresponding home shares from 26 developed and emerging countries/exchanges around the globe; the second set includes large, mid, and small-cap portfolios of ADRs. Our results based on bivariate VAR methodology indicate bidirectional causality for ADR and their corresponding home shares portfolios for almost all our sample countries/exchanges. We compute daily returns from closing prices and volatility and spreads using daily high and low prices. Our results on large, mid, and small-cap portfolio returns indicate a central role for small-cap portfolios to lead returns in large and mid-cap portfolios. We further investigate these portfolio returns, volatility, and spread.

I. Introduction

We empirically investigate two comprehensive hypotheses on the time dependence of auto and cross covariances of three portfolio performance measures, namely, returns, volatilities, and spreads of synthetic portfolios of ADRs and their underlying securities from 26 global exchanges.

In order to test our first hypothesis- inter market time varying spillover hypothesis, we create two matching portfolios of ADRs and their corresponding home shares from each country/exchange and then in a VAR setup determine the lead lag relations in return, volatility, and spread across each matched pair of portfolios of ADRs and their corresponding underlying home securities. We draw motivation from two distinct lines of documented evidence on return and volatility spillover. First, Lo and McKinlay (1990) and Chordia and Swaminathan (2000) confirm that large and liquid firms lead small and less liquid firms returns; second, Diebold and Yilmaz (2009) observe that “returns spillovers display an increasing trend but no bursts, whereas

volatility spillovers display no trend but clear bursts.” The majority of ADRs are smaller and less liquid than their underlying home shares and hence we may expect home share portfolios to lead ADR portfolios returns. We conduct two causality tests, Granger causality and Instantaneous causality to determine a consistent lead lag relation, if any, between ADR and home shares portfolios returns, volatilities, and spreads. Following up on Diebold and Yilmaz (2009), we use a variance decomposition technique and a rolling sample to compute a time varying spillover index that tracks how the respective auto and cross covariances of ADR and home security portfolios’ performances commove over time, in particular through booms and bursts in the US market.

Much of the empirical evidence on spillover is based on data on market indexes and address questions about whether and how volatility spillover and to a lesser extent return spillover change over time and are predicated on international market integration and spread of financial contagion across markets during crises and non-crises periods (Bekaert et al [2014]).

Understandably, many of those studies analyze long time series data for the purpose of identifying shifting trends. On the contrary, there are also a few limited horizon studies, which provide scattered evidence on both return and volatility spillovers across dually listed securities and securities baskets, and sector portfolios (Chen et al. [2009], Krause and Tse [2013], Guitierrez et al. [2009], Poshakwale [2008], Jaiswal-Dale and Jitendranathan [2009]). We are not aware of any such studies on inter portfolio/market spillover of spread.

Our second hypothesis builds on the empirical evidence that stock returns are serially correlated and further it’s possible to exploit such time series dependence in stock returns to derive excess returns from carefully constructed portfolios of those stocks (Lo and McKinlay [1990]). In subsequent studies, Chordia and Swaminathan (2000), Hou (2007), DeMiguel et al.

(2014) confirm the lead lag evidence in returns with respect to liquidity, industry diffusion, value vs. growth, and momentum as a proxy for feedback trading. We construct three size stratified portfolios, small, mid, and large cap from our sample of ADRs only and conduct a VAR analysis to test whether the lead lag return cross correlation between large and small firms observed in Lo and McKinlay (1990) holds consistently for similar portfolios from developed and emerging markets. We conduct VAR testing for returns on developed vs. emerging markets portfolios with additional control based on trading volume, turnover, industry diffusion, and feedback trading. We also conduct similar VAR analyses of volatility and spread. Finally, partially driven by Chordia et al. (2011), we conduct a cointegration analysis to determine whether return, volatility, and spread commove due to a common ‘information’ vector that initiates price movements affecting simultaneously return, volatility, and spread.

We compute portfolio performance measures based on daily closing as well as high and low prices and use a VAR setup for our empirical investigation. The VAR framework allows us to simultaneously estimate multiple lags of auto and cross covariance parameters associated with ADR and home portfolios returns, volatilities, and spreads over time. We use daily close to close prices for computing returns and high and low prices to compute volatility and spread. Daily high and low prices are preferable to opening and closing prices for capturing transient intraday shocks, which may not be revealed in opening and/or closing prices. Indeed, in terms of distributional properties, the superiority of a range based volatility measure over other alternative measures is well documented in the literature and finally, recently proposed daily high low prices based spread measurement techniques allow spreads to be computed over very long time. We

follow Parkinson (1980) and Corwin and Schultz (2012) to compute volatility and spread respectively using daily high and low prices.¹

Results from bivariate VAR tests between ADR and matched home shares portfolios indicate scattered auto correlation but highly significant multi-lag cross dependence for all three variables- return, spread, and volatility. Granger causality tests indicate strong bidirectional causality between ADR (NYSE) and home shares (primary listing exchange) portfolios for return, volatility, and spread for most countries/exchanges;

II. Data and Empirics on ADR and Home Portfolios

II.A. Data and Sample Statistics

We begin with a complete list of 525 foreign listings from 46 countries on the New York Stock Exchange (hereafter NYSE) and NYSE Market as of December 31, 2012. Among those foreign listings, 259 are foreign equities (common and preferred) with primary listing in an exchange located outside the US, and traded at the NYSE as ADRs (including Global Shares and NY Registry Shares).² Table 1 provides a breakdown of our sample of 186 ADRs and their corresponding home country shares listed in domestic exchanges in 16 developed and 14 emerging markets belonging to MSCI Developed and Emerging Markets Indexes respectively.

¹ Since Parkinson (1980) and Garman and Klass (1980) many published works including Brandt and Diebold (2006) have noted the efficiency of a range, high minus low price, based measure of volatility. For a literature review on range based volatility measures, please refer to Bali and Weinbaum (2005) and Chou et al. (2009). The use of high and low prices in determining spread is more recent and originates from the lack of available intraday order/quotes level data in many exchanges necessary for estimating bid ask spread. Recently, besides Corwin and Schultz (2012), Holden (2009) and Chung and Zhang (2013) use range based empirical measures of spread.

² Global and NY registry shares are for securities from a couple of countries like Netherlands, which due to the local exchange restrictions cannot sale shares in another currency.

Table 2 contains our sample ADRs and their underlying home securities with other relevant details. After eliminating securities with missing data (23 securities pairs marked with asterisks are either preferred shares, do not have any data, or have limited data and hence excluded from the empirical analysis), we have a total of 163 pairs of ADR and their corresponding home shares from 26 developed and emerging market countries distributed as follows: Australia (6), Belgium (2), Brazil (16), Chile (10), China (12), Denmark (1), Finland (1), France (7), Germany (3), India (6), Indonesia (1), Ireland (3), Israel (2), Italy (4), Japan (15), Luxembourg (3), Korea (7), Mexico (12), Netherlands (5), Norway (1), Philippines (1), Portugal (1), Spain (3), South Africa (6), Switzerland (5), Taiwan (6), and UK (23).³

We obtain daily prices (high, low, open, and close), value of traded shares (turnover), and market value of equity for those 163 ADRs and their underlying home securities listed on the respective domestic Stock Exchanges from Bloomberg for a period of approximately eight to eighteen years, from the initial trading day for each security beginning on or after January 1994. Hence, the sample period for each pair of ADR and its underlying security begins from their respective initial trading dates from/after January 1, 1994 and ends on September 20, 2013. We further obtain prices, turnover (value of traded shares), and market value of equities for those 26 developed and emerging markets in which our sample securities are primarily listed.

We compute daily returns as the difference between log closing prices for two consecutive trading days. We also compute security returns, $r_{i,t}$ for the i th ADR or its underlying home

³ MSCI Emerging Markets Index spans across five continents, Asia, Europe, North America, South America, and Africa and includes the following 21 countries- Brazil, Chile, China, Colombia, Czech Republic, Egypt, Hungary, India, Indonesia, Korea, Malaysia, Mexico, Morocco, Peru, Philippines, Poland, Russia, South Africa, Taiwan, Thailand, and Turkey. As of January 31, 2013, seven of those countries, Czech Republic, Egypt, Hungary, Malaysia, Morocco, Poland, and Thailand do not have any ADRs listed on US exchanges. Further 23 ADRs/home country shares are either preferred shares and/or do not have the necessary data and hence are excluded. Included in those 23 securities are two each from Colombia and Russia and one each from Peru and Turkey, which denote their entire ADR listings respectively and are therefore completely eliminated from the sample due to missing data. Hence our final sample consists of 78 securities from 10 emerging markets.

security on day t as $\ln\left(\frac{P_t^H + P_t^L}{P_{t-1}^H + P_{t-1}^L}\right)$ from daily midpoints between high and low prices (midpoint return later) in order to explicitly account for large intraday variances, which may not be manifested in opening and closing prices on a day or during an interval, e.g., over 24 hours. For comparison purpose, we also compute day return from day open and close prices, and overnight (denotes non-trading hours) return from day close to next day open prices for each ADR and its underlying home security. Next, using appropriate time varying weights, we compute daily, midpoint, and overnight returns on equally weighted portfolios of ADRs and their corresponding home securities for each trading day, t in our sample, as below.

$$\tilde{r}_{A,d,t} = \sum_{i=1..A} w_i r_{i,d,t} \text{ and } \tilde{r}_{A,n,t} = \sum_{i=1..A} w_i r_{i,n,t} \dots\dots\dots(1a, 1b)$$

$$\tilde{r}_{H,d,t} = \sum_{i=1..H} w_i r_{i,d,t} \text{ and } \tilde{r}_{H,n,t} = \sum_{i=1..H} w_i r_{i,n,t} \dots\dots\dots(2a, 2b)$$

In equations 1 and 2 above, $r_{i,t}^A$ denotes continuously compounded return of the i th ADR for the t th day; $r_{i,t}^H$ denotes continuously compounded return of the underlying security corresponding to the i th ADR for the t th day; $r_{i,n}^A$ denotes continuously compounded return of the i th ADR for the n th overnight preceding the t th day; $r_{i,n}^H$ denotes continuously compounded return of the underlying security corresponding to the i th ADR for the n th overnight preceding the t th day; $\widetilde{r_{A,d,t}}$ and $\widetilde{r_{A,n,t}}$ are daily and overnight returns for the t th day and the preceding overnight for an ADR portfolio while $\widetilde{r_{H,d,t}}$ and $\widetilde{r_{H,n,t}}$ are similar returns on the corresponding home shares portfolio respectively.

Table 3A provides descriptive statistics on intraday midpoint returns for each ADR and its underlying home share listed at a domestic stock exchange. In addition, Table 3B contains means and standard deviations of daily (close to next day close), day (open to close), and

overnight (close to next day open) returns for equally weighted portfolios of ADRs and their corresponding home securities. Table 3B also contains correlations between pairs of midpoint and daily, midpoint and day, and midpoint and overnight returns.

The descriptive statistics indicate asymmetric differences between means of trading day and daily returns for both ADRs and their corresponding home securities, albeit some of the differences seem to be due to the varying length of each security's time series. For example, during the sample period, while most ADRs and their underlying home shares show positive mean daily returns, most securities show negative mean day returns during the sample period. Standard deviations of daily returns are consistently higher than those of day returns. Figure 1 (not included in the current draft) shows distribution of risk denoted by standard deviation against mean return for daily and day return respectively, it indicates higher level of dispersion in daily return distribution.

Table 3C reports first order autocorrelations for each of our computed return series, midpoint, daily, day, and overnight returns for both ADR and home shares.

Table 3D (not included in the current draft) contain parameter estimates for two sets of joint mean variance equality tests for the sample ADRs and their corresponding home shares' portfolio returns. First, Panel A contains the test statistic for the null hypothesis that the means and variances of daily and midpoint returns are equal; next, Panel B tests the null hypothesis that the means and variances of ADR and home shares portfolios are equal. As expected, the equality of distributions based on means and variances between midpoint daily returns is rejected for each ADR and its underlying home security; the equality of distributions between ADRs and their corresponding home securities returns are rejected for selected portfolios. Such differences in the distributional properties of daily and overnight returns bias the results of those studies which

combine day and daily returns to account for large time differences between markets, which do not have overlapping market open hours.

Next, we compute volatility ($v_{i,t}$) and spread ($s_{i,t}$) for each security i on day t using their daily high and low prices as in equations 3a through 3d below and then continue on to compute volatility and spread for ADR and underlying home portfolios respectively. The computations of portfolio volatility and spread is complicated by the fact that portfolio volatility needs to include covariance terms and the weights for portfolio spread are to be determined by volume/turnover.

$$v_{i,t} = \sigma_{i,t}^2 = \ln\left(\frac{p_t^H}{p_t^L}\right)^2 * \left(\frac{1}{4\ln(2)}\right) \dots (3a)$$

$$s_{i,t} = \frac{2(e^\alpha - 1)}{1 + e^\alpha} \dots (3b)$$

$$\alpha = \frac{\sqrt{2\beta} - \sqrt{\beta}}{3 - 2\sqrt{2}} - \sqrt{\frac{\gamma}{3 - 2\sqrt{2}}} \dots (3c)$$

$$\beta = E\left[\sum_{j=0}^1 \ln\left(\frac{p_{0,t+j}^H}{p_{0,t+j}^L}\right)\right] \text{ and } \gamma = \left[\ln\left(\frac{p_{0,t,t+1}^H}{p_{0,t,t+1}^L}\right)\right]^2 \dots (3d).$$

We compute volatility and spread as per equations 3a through 3d following Parkinson (1980) and Corwin and Schultz (2012) respectively. While absolute returns, squared returns, and standard deviation of returns are the most common estimates of volatility, since Parkinson (1980) and Garman and Klass (1980) range based estimates of volatility have frequently found their way into the finance literature (Bali and Weinbaum [2005], Chou et al [2009]). On the contrary, only recently Holden (2010), Corwin and Schultz (2012), Chung (2010) have used high and low prices to estimate spread thus circumventing the burden of intraday trade and quotes data for

estimating spread. Table 4 presents descriptive statistics on volatility and spread computed as above.

II.B. Empirical Models and Results

We empirically determine how auto and cross portfolio lag returns impact portfolio returns in the following two steps. First, we choose an appropriate model for deseasonalizing returns as in equations 4a and 4b below and follow up with a bivariate VAR model as in equations 5a and 5b hereunder.

$$\tilde{r}_{A,t} = \alpha_A + \sum_{m=1}^{11} \varphi_m \delta_m + \lambda \tilde{r}_{A,n} + \sum_{d=1}^4 \chi_d \delta_d + \varepsilon_{A,t} \dots\dots\dots(4a)$$

$$\tilde{r}_{H,t} = \alpha_H + \sum_{m=1}^{11} \varphi_m \delta_m + \lambda \tilde{r}_{H,n} + \sum_{d=1}^4 \chi_d \delta_d + \varepsilon_{H,t} \dots\dots\dots(4b)$$

where δ_d denotes day of the week dummy; δ_m denotes first/last day of month in a year dummy.

$$\varepsilon_{A,t} = \alpha_A + \sum_{j=1}^J \alpha_j^A \varepsilon_{A,t-j} + \sum_{k=1}^K \beta_k^A \varepsilon_{H,t-k} + e_{A,t} \dots\dots\dots(5a)$$

$$\varepsilon_{H,t} = \alpha_H + \sum_{j=1}^J \alpha_j^H \varepsilon_{H,t-j} + \sum_{k=1}^K \beta_k^H \varepsilon_{A,t-k} + e_{H,t} \dots\dots\dots(5b)$$

We find scattered and limited evidence of significant first/last day of month but no day-of-the week effects in returns. For the sake of brevity, we do not report results from equations 4a and 4b; nevertheless, we use the residuals from those equations considering only the significant parameters. Tables 5A and 5B contain the parameter estimates and their corresponding t-stats of the above bivariate VAR model for ADR and home portfolio returns as in equations 5a and 5b.

In each table, the coefficients α_1 and $\Sigma\alpha_5$ relate to lag 1 and lag 1-5 of auto covariances while β_1 and $\Sigma\beta_5$ correspond to lag 1 and lag 1-5 of cross covariances of ADR and home shares portfolio returns. We include each portfolio's overnight returns as an exogenous variable in the above VAR models.

The parameter estimates for VAR for portfolio returns are as follows. For ADR return, the coefficients for the one lag ADR return (α_1) are all positive and significant at 1% level with the exception of those of Israel and Mexico, which are insignificant (at greater than 10% level). The coefficients associated with the sum of lag 1-5 of ADR returns show reversals (the coefficients become less positive, zero, or even negative) and higher standard errors (11 insignificant including Mexico, 8 significant at 1%, and 2 significant at 10% level respectively). Regarding one lag cross covariance, all home shares portfolio returns except those of Brazil and Chile show significant, negative coefficients. Further, the coefficients with respect to sum of lag 1-5 home share portfolio returns show reversals while 13 ADR portfolios including Chile exhibit insignificant statistics. Overnight returns show positive and significant at 1% level impact (coefficient around 1.00) on ADR returns and adjusted R^2 for the sample ADR portfolios ranges between 60 and 80 percent.

We find qualitatively almost similar results for returns on home share portfolios. The coefficients corresponding to one lag home share portfolio returns are significantly negative (21), significantly positive (3) and insignificant (Israel and Taiwan) while those related to one lag ADR portfolio returns are significantly positive (Mexico insignificant). The coefficients associated with higher lags of both home portfolio and ADR portfolio returns show reversals and higher standard errors; coefficient estimates corresponding to overnight returns range from 0-

99% with large variations in standard errors causing t-stats to span between -1.91 and 94.68. Adjusted R^2 ranges from 15% (Italy) to 75% (Japan).

We take note of the uniform and consistent positive and negative first order auto covariance estimates for ADR and home share portfolios respectively. The negative auto covariances for home share returns are particularly surprising given that in Table 3C, we find auto correlations for home shares are all positive. This sign change in the VAR estimates indicates a definite role for the cross covariance affecting the auto covariances and hence strengthens the argument for conditional over unconditional estimates of covariance as determinants of returns.

II.C. VAR for Volatility and Spread

$$v_{A,t} = \alpha_{v,A} + \sum_{j=1}^J \alpha_j^A v_{A,t-j} + \sum_{k=1}^K \beta_k^A v_{H,t-k} + e_{v,A,t} \dots\dots\dots(6a)$$

$$v_{H,t} = \alpha_{v,H} + \sum_{j=1}^J \alpha_j^A v_{H,t-j} + \sum_{k=1}^K \beta_k^A v_{A,t-k} + e_{v,H,t} \dots\dots\dots(6b)$$

Similarly,

$$s_{A,t} = \alpha_{s,A} + \sum_{j=1}^J \alpha_j^A s_{A,t-j} + \sum_{k=1}^K \beta_k^A s_{H,t-k} + e_{s,A,t} \dots\dots\dots(6c)$$

$$s_{H,t} = \alpha_{s,H} + \sum_{j=1}^J \alpha_j^H s_{H,t-j} + \sum_{k=1}^K \beta_k^A s_{A,t-k} + e_{s,H,t} \dots\dots\dots(6d)$$

Tables 6A-6B contain the parameter estimates of the above bivariate VAR models, equations 6a and 6b for ADR and home shares portfolio variance respectively; Tables 6C-6D contain similar estimates related to the equations 6c and 6d for spreads for ADR and home shares portfolios respectively. We include return variances for ADR and home share portfolios

measured as ε^2 estimated from equations 4a and 4b substituting daily midpoint returns with returns computed from close to close prices as exogenous variables in equation 6a and 6b respectively. Similarly, we include return variances computed from high low prices as in equation 3a as exogenous variables in the VAR models for spread in equations 6c and 6d.

We make the following observations regarding the estimated parameters for equation 6a and 6b. First, the first lag of auto covariance is mostly positive (for ADR variance, Indonesia, Israel, Italy are insignificant; for home share variance, Israel is insignificant); second, the sum of lag auto covariance is also significantly positive or insignificant (for ADR variance, Italy and for home share variance, Ireland insignificant) for all ADR and home share portfolios; third, the first lag cross covariance is significantly positive for 18 (23) ADR (home share) portfolios; the sum of 5 lag cross covariance is significantly positive for 14 (13) ADR (home share) portfolios; close to close return volatility (coefficient range 0.39-1.75 for ADR volatility and 0.02-0.56 for home share volatility) is significantly positive; adjusted R^2 ranges between 0.04 (Italy) and 0.82 (Netherlands) for ADR volatility and between 0.13 (Israel) and 0.71 (Belgium).

Similar observations are made for the estimated parameters of ADR and home share spreads. The first order and the sum of the auto regressive coefficients are highly significant for both ADR and home share spreads; on the contrary, while the number of significant first order (sum of) lag(s) of the cross variable is 12 (19) for ADR spread, it is 26 (15) for home shares spread. Volatility is significant for all ADR and home share portfolios and the adjusted R^2 ranges between 58% and 86% for both ADR and home share portfolios. We summarize the VAR results for return, volatility, and spread in Table 7 below.

Table 7: Summary of significant (+/-) and insignificant (0) bivariate VAR parameter estimates

	Auto			Cross		
	1 lag	1-5 lag	Comments on parameter estimates	1 lag	1-5 lag	Comments on parameter estimates
Return-ADR	24 +, 2 0	7 +, 8 -, 11 0		24 -, 2 0	3+, 10-, 13 0	
Return-Home	3 +, 21 -, 2 0	17 -, 9 0		25 +, 1 0	17 +, 1 -, 8 0	
Volatility-ADR	23 +, 3 0	25 +, 1 0		18 +, 8 0	13 +, 1 -, 12 0	
Volatility-Home	25 +, 1 0	25 +, 1 0		23 +, 3 0	9 +, 4 -, 13 0	
Spread-ADR	26 +	26 +		9 +, 2-, 15 0	18 +, 1 -, 7 0	
Spread-Home	26 +	26 +		23 +, 1 -, 2 0	5 +, 10 -, 11 0	

II.D. Causality, Variance Decomposition, and Sub-period Analysis

We test the following null hypotheses with regards to the VAR models given in equations 5a-5b for returns and equations 6a-6d for volatility and spread respectively.

1. H_0 : Home equity returns Granger-cause ADR returns
2. H_0 : ADR returns Granger-cause Home equity returns
3. H_0 : No instantaneous causality exists between ADR and Home equity returns
4. H_0 : Home equity volatility Granger-causes ADR volatility
5. H_0 : ADR volatility Granger-causes Home equity volatility
6. H_0 : No instantaneous causality exists between ADR and Home equity volatility
7. H_0 : Home equity spread Granger-causes ADR spread
8. H_0 : ADR spread Granger-causes Home equity spread
9. H_0 : No instantaneous causality exists between ADR and Home equity spread

Table 8 reports F-tests for Granger causality and Chi-square tests for instantaneous causality corresponding to the nine null hypotheses above. The causality tests statistics almost

uniformly and consistently reject the null hypotheses that returns, volatilities, and spreads of ADR and home shares portfolios do not instantaneously and Granger cause returns, volatilities, and spreads of each other. The only exceptions are Australia: the test statistics fail to reject the null hypothesis of home equity returns do not Granger cause ADR returns; Israel: the test statistics fail to reject the null hypotheses of home equity volatility and spread do not Granger cause ADR volatility and spread respectively; and Italy: the test statistics fail to reject the null hypothesis of home equity returns do not Granger cause ADR returns. The bi-directional causality contradicts findings on the documented mostly one way return and volatility transmission from the US to international markets.

Our sample period spans over a range of eight to 21 years during which the US market and the respective home markets have passed through multiple phases of bull and bear markets (identification strategy NBER). Further, while the Wald test results in Table 8 strongly support Instantaneous causality for returns, volatilities, and spreads, 3 countries portfolios fail to reject home to ADR lead lag relations. Those and the hugely varying parameter estimates in Table 7 suggest that there may be large time varying components in the cross covariances between returns, volatilities, and spreads of ADR and home securities, which impact markets asymmetrically over time and hence warrant additional probing. In order to test for time varying cross covariance, we run the above VAR analyses on a 500 days rolling sample.

III. Common Drivers of Lead Lag in ADR Portfolio Performance Measures

In this section, we investigate whether and how size differences may explain the lead lag relation evident in return, spread, and volatility and whether size differences trump country level

differences in explaining lead lag effects. Our base model is that of Lo and McKinlay (1990); however, we expand the model by including additional factors besides size, as liquidity, momentum, industry diffusion, and feedback trading in order to determine whether portfolio specific differences edge over country level differences in explaining lead lag relations in portfolio performance denoted by returns, volatility, and spread. For this analysis, we focus only on the ADR portfolio since the majority of individual home country portfolios have too few securities to create sub-portfolios based on security specific differences, for example, liquidity.

Our empirical setup is as follows. First, we sort the ADRs in decreasing order of market capitalization and further construct three portfolios, large, mid, and small cap with those stocks. Second, for each portfolio P_{it} (where $i=\{large, mid, small\}$, $t=\{1,T\}$) we compute return, volatility, and spread using high and low prices as above and in a VAR setup test the null hypotheses that return, volatility, and spread do not spillover from one portfolio (large, mid, or small cap) to another. Third, we include emerging vs. developed and region dummies in each VAR equation to test for spillover in the presence of country and market specific effects.

We begin with testing the return spillover hypotheses. Table 9 contain select descriptive statistics on large, mid, and small-cap portfolio returns along with difference in means tests for portfolio returns. While mean returns for large and mid-cap portfolios are similar, mean returns for small-cap portfolios seem to lagging behind; however, only mid to small cap portfolios show significant (at less than 10% level) difference in means. Notably though, in sub-sample analysis, both large to small cap portfolios and mid to small-cap portfolios show statistically significant differences only during 1995-96, which indicate time variation in mean portfolio returns.

Table 10 contains parameter estimates and their corresponding t-statistics for three bivariate VAR models involving large, mid, and small-cap portfolio returns. Further, we report

appropriate F-statistics pertaining to Granger causality tests. The test statistics suggest a complete reversal of Lo and McKinlay (1990) results- small firms' returns Granger cause both mid and large-cap returns while large cap returns do lead small-cap but not mid-cap returns. We continue investigating the lead lag effects of ADR portfolios.

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Table 1: Sample breakdown into countries included in MSCI ACWI index, which is composed of 23 developed and 23 emerging markets indexes as below.

	Total	MSCI	Sample
Nos. of non-US issuers (countries) listed on NYSE/NYSE Market as of Dec 2012	525 (46)		
Nos. of ADRs (countries with >=1 ADR) listed on NYSE as of Dec 2012	259 (33)		
ADRs (non-US Countries) included in MSCI Developed Markets		81 (16)	81 (16)
Australia		6	6
Belgium		2	2
Denmark		1	1
Finland		1	1
France		7	7
Germany (1 preferred)		4	3
Ireland		3	3
Israel		2	2
Italy		4	4
Japan		15	15
Netherlands (3 NY Registry Shares)		5	2
Norway		1	1
Portugal		1	1
Spain		3	3
Switzerland (1 NY Registry Shares)		5	4
UK		23	23
Luxembourg*		3	
ADRs (Countries) included in MSCI Emerging markets		148 (15)	76 (11)
Brazil (1 preferred)		25	16
Chile (1 preferred)		12	10
China		60	12
Colombia		2	
Greece		1	1
India		8	6
Indonesia		1	1
Mexico (1 Global Share)		14	12
Peru		2	
Philippines		1	1
Russia		2	
South Africa		7	6
South Korea		8	7
Taiwan		5	5
Turkey		1	
Argentina*		10	

The **MSCI ACWI** consists of 23 developed (DM) and 23 emerging market (EM) country indexes. The developed market country indexes included are: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, the United Kingdom and the United States. The emerging market country indexes included are: Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Korea, Malaysia, Mexico, Peru, Philippines, Poland, Qatar, Russia, South Africa, Taiwan, Thailand, Turkey (included 6/14) and United Arab Emirates. Luxembourg and Argentina are not included in the MSCI ACWI index.

Table 2: Sample description of 101 Emerging Markets ADRs listed on NYSE system and their underlying shares listed in the respective home countries

Ticker	Name	Primary Exchange Country	Industry	Exchange Ratio	Date of ADR [primary] listing
AWC (AWCMY)	Alumina Ltd	Australia			1/90
BHP (BHP)	BHP Billiton	Australia			5/87
JHX (JHX)	James Hardie Industries	Australia			10/01
OBTLY (OEC)	Orbital Corp Ltd	Australia			8/11
SSN (SSN)	Samson Oil and Gas	Australia			1/08
SMS ()**	Sims Group Ltd	Australia			3/08
WBK (WBC)	Westpac Banking Corp	Australia			3/89
BUD (ABI)	Anheuser Busch	Belgium			9/09
DEG (DELB)	Delhaize Group	Belgium			4/01
NVO (NOVOB)	Novo-Nordisk	Denmark			7/81
NOK (NOK1V)	Nokia Corp	Finland			7/94
ALU (ALU)	Alcatel-Lucent	France			5/92
CGG ()**	CGG	France			5/97
ORAN (ORA)	Orange SA	France			10/97
SNY (SAN)	Sanofi SA	France			7/02
TOT (FP)	Total SA	France			10/91
VE (VIE)	Veolia Environment	France			10/01
FMS (FME)	Fresenius AG	Germany			9/96
SAP (SAP)	SAP SE	Germany			8/98
SI (SIEGY)*	Siemens AG	Germany			3/01
CRH (CRH)	CRH PLC	Ireland			3/06
FLY ()**	Fly Leasing Ltd	Ireland			9/07
ELN (ELN)*	Elan Corp PLC	Ireland			1/95
IRE (BKIR)	Bank of Ireland	Ireland			9/96
BSI (BSI)	Blue-Square	Israel			8/96
TEVA (TEVA)	Teva Pharmaceuticals	Israel			5/12

Ticker	Name	Primary Exchange Country	Industry	Exchange Ratio	Date of ADR [primary] listing
E (ENI)	ENI SpA	Italy			11/95
LUX (LUX)	Luxotica Group SpA	Italy			1/90
NTZ ()**	Natuzzi SpA	Italy			5/93
TI (TIT)	Telephone Italia	Italy			7/95
ATE (6857)	Advantest Corp	Japan			9/01
CAJ (7751)	Canon Inc	Japan			9/00
HMC (7267)	Honda Motor Corp	Japan			2/77
KNM (9766)	Konami Corp	Japan			9/02
KUBTY (6326)*	Kubota Corp	Japan			11/76
KYO (6971)	Kyocera Corp	Japan			5/80
MTU (8306)	Mitsubishi Holdings	Japan			9/89
MFG (8411)	Mizuho Financial Grp	Japan			11/06
NJ (6594)	Nidec Corp	Japan			9/01
NTT (9432)	Nippon Tel	Japan			9/94
NMR (8604)	Nomura Holdings	Japan			12/01
DCM (9437)	NTT DoCoMo	Japan			3/02
IX (8591)	Orix Corp	Japan			9/98
PC (6752)*	Panasonic Corp	Japan			12/71
SNE (6758)	Sony Corp	Japan			9/70
SMFG (8316)	Sumitomo Mitsui Fin Group	Japan			11/10
TM (7203)	Toyota Motors	Japan			9/99
AEG (AGN)	AEGON (New York Registry)	Netherlands			11/91
ING (INGA)	ING	Netherlands			6/97
PHG (PHIA)	Phillips Electronics	Netherlands			4/87
ENL (REN)	Reed Elsevier	Netherlands			10/94
UN (UNA)	Unilever (New York Registry)	Netherlands			12/61
STO (STL)	Statoil	Norway			6/01
PT (PTC)	Portugal Telecom	Portugal			6/95
BBVA (BBVA)	Banco Bilbao S.A.	Spain			12/88
SAN (SAN)	Banco Santander S.A.	Spain			7/87

Ticker	Name	Primary Exchange Country	Industry	Exchange Ratio	Date of ADR [primary] listing
TEF (TEF)	Tefonica S.A.	Spain			6/87
ABB (ABBN)	ABB	Switzerland			4/01
CS (CSGN)	Credit Suisse	Switzerland			9/01
NVS (NOVN)	Novartis AG	Switzerland			5/00
STM (STM)	STMicroelectronics (NY Registry)	Switzerland			12/94
SYT (SYNN)	Syngenta AG	Switzerland			11/00
AZN (AZN)	Astrazane PLC	UK			5/93
AV (AV)	Aviva PLC	UK			10/09
BCS (BARC)	Barclays	UK			9/86
BBL (BLT)	BHP Billiton	UK			6/03
BP (BP)	British Petroleum	UK			3/70
BTI (BATS)	British American Tobacco	UK			3/28
BT (BT/A)	BT Group	UK			12/84
CUK (CCL)	Carnival PLC	UK			10/00
DEO (DGE)	Diageo PLC	UK			12/97
GSK (GSK)	GlaxoSmithKline	UK			6/87
HBC (HSBA)	HSBC Holdings	UK			7/99
IHG (IHG)	Intercontinental Hotels	UK			3/03
LYG (LLOY)	Lloyds Banking	UK			11/01
NGG (NG)	National Grid	UK			10/99
PSO (PSON)	Pearson plc	UK			9/00
PUK (PRU)	Prudential Insurance	UK			6/00
RUK (REL)	Reed Elsevier plc	UK			10/94
RIO (RIO)	Rio Tinto plc	UK			6/90
RDSA ()**	Royal Dutch Shell plc	UK			7/05
SNN ()**	Smith and Nephew plc	UK			11/99
RBS ()**	Royal Bank of Scotland Grp plc	UK			10/07
UL ()**	Unilever	UK			12/61
BBD [BBDC3]	Banco Bradesco S.A	Brazil		1:1	11/01
BSBR [SANB11]	Banco Santander S.A	Brazil		1:1	10/09

Ticker	Name	Primary Exchange Country	Industry	Exchange Ratio	Date of ADR [primary] listing
LND [AGRO3]*	BrasilAgro Companhia	Brazil		1:1	11/12
OIBR [OIBR3]	Oi S.A	Brazil		1:1	11/01
BAK [BRKM3]*	Braskem S.A	Brazil		1:2	12/98
BRFS [BRFS3]*	BRF- Brasil Foods S.A	Brazil		1:1	10/00
EBR [ELET3]	Centrais Electricas- Electrobras	Brazil		1:1	10/08
CBD [PCAR4]*	Companhia Brasileira	Brazil		1:1	5/97
ABV [AMBV3]*	AmBev	Brazil		1:1	6/97
SBS [SBSP3]	Sabesp	Brazil		1:1	5/02
CIG [CMIG3]	Cemig	Brazil		1:1	9/01
ELP [CPLE3]*	COPEL	Brazil		1:1	7/97
SID [CSNA3]	CSN	Brazil		1:1	11/97
CPL [CPFE3]	CPFL	Brazil		1:2	9/04
ERJ [EMBR3]	Embraer S.A	Brazil		1:4	7/00
FBR [FIBR3]	Fibria Celulose S.A	Brazil		1:1	4/00
GFA [GFSA3]	Gafisa S.A	Brazil		1:2	3/07
GGB [GGBR3]*	Gerdau S.A	Brazil		1:1	3/99
GOL ^{PRF} GOLL4]*	GOL	Brazil		1:1	6/04
ITUB [ITUB3]*	Itau Unibanco S.A	Brazil		1:1	2/02
PBR [PETR3]	PETROBRAS	Brazil		1:2	8/00
VIV [VIVT]	Telefonica Brasil S.A	Brazil		1:1	11/98
TSU [TIMP3]	TIM S.A	Brazil		1:5	11/98
UGP [UGPA3]	Ultrapar S.A	Brazil		1:1	10/99
VALE [VALE3]	VALE S.A	Brazil		1:1	6/00
PVD [PROVIDA]	A.F.P Provida S.A	Chile		1:15	11/94
BCH [CHILE]	Banco de Chile	Chile		1:600	1/02
BSAC [BSAN]	Banco Santander-Chile	Chile		1:400	1/97
CNCO ()**	Cencosud SA	Chile			6/12
CCU [CCU]	Compania .. Unidas S.A	Chile		1:2	3/99
BCA [CORPBANC]	Corpbanca	Chile		1:1500	11/04
AKO/A ^P [ANDINAA]*	Embotalladora Andina S.A	Chile		1:6	7/94

Ticker	Name	Primary Exchange Country	Industry	Exchange Ratio	Date of ADR [primary] listing
EOC [ENDESA]	Empresa Nacional de S.A	Chile		1:30	7/94
ENI [ENERSIS]	Enersis S.A	Chile		1:50	10/93 [3/13]
LFL [LAN]	LAN Airlines S.A	Chile		1:1	11/97
SQM [SQM/B]	SQM	Chile		1:1	9/93 [10/95]
VCO [CONCHA]	Vina Concha ...S.A	Chile		1:20	10/94
ACH [2600]	Aluminium Corp of China Ltd	China Hshares		1:25	12/01 [12/01]
CEA [0670]	China Eastern Airlines Corp Ltd	China Hshares		1:50	01/97 [02/97]
LFC [2628]	China Life Insurance Co Ltd	China Hshares		1:15	12/03 [12/03]
CHA [0728]	China Telecom Corp Ltd	China Hshares		1:100	11/02 [11/02]
CHL [0941]**	China Mobile	China Hshares			
SNP [0386]	China Petroleum & Chem. Corp	China Hshares		1:100	10/00 [10/00]
ZNH [1055]	China Southern Airlines Co Ltd	China Hshares		1:50	07/97 [07/97]
CEO [0883]**	CNOOC Ltd	China Hshares		1:100	02/01 [02/01]
GSH [0525]	Guangshen Railway Co Ltd	China Hshares		1:50	05/96 [05/96]
HNP [0902]**	Huaneng Power International Inc	China Hshares		1:40	08/03 [01/98]
PTR [0857]	PetroChina Co Ltd	China Hshares		1:100	03/00 [04/00]
SMI [981]	Semiconductor Mfg. Intl. Corp	China Hshares		1:50	03/04 [03/04]
SHI [0338]	Sinopec Shanghai Petroch. Co Ltd	China Hshares		1:100	07/93 [07/93]
YZC [1171]	Yanzhou Coal Mining Co Ltd	China Hshares		1:10	03/98 [03/98]
CIB ^{PR} [PFBCOLO]*	BanColombia S.A	Colombia		1:4	7/95
EC*	Ecopetrol S.A	Colombia		1:20	9/08 [11/07]
NBG []**	National Bank of Greece	Greece			10/99
INFY [INFO]	Infosys Ltd	India		1:1	12/12
IBN [ICICIBC]	ICICI Bank Ltd	India		1:2	03/00
TTM [TTMT]	Tata Motors Ltd	India		1:5	09/04
SSLT [STLT]**	Sesa Sterlite Ltd	India		1:4	06/07
WIT [WPRO]	Wipro Ltd	India		1:1	10/00
HDB [HDFCB]	HDFC Bank Ltd	India		1:3	07/01
RDY [DRRD]	Dr. Reddy's Laboratories Ltd	India		1:1	04/01
WNS []**	WNS Holdings Ltd	India			07/06

Ticker	Name	Primary Exchange Country	Industry	Exchange Ratio	Date of ADR [primary] listing
TLK [TLKM]	Telekom Indonesia Pers Tbk PT	Indonesia		1:40	11/95
LPL [034220]	LG Display Co. Ltd	South Korea		1:0.5	07/04
SKM [017670]	SK Telecom Co. Ltd	South Korea		1:0.11	06/96
KEP [015760]	Korea Electric Power Co	South Korea		1:0.5	10/94
PKX [005490]	POSCO	South Korea		1:0.25	10/94
KT [030200]**	KT Corp	South Korea		1:0.5	05/99
SHG [055550]	Sinhan Financial Group Co. Ltd	South Korea		1:1	09/03
KB [105560]	KB Financial Group Inc	South Korea		1:1	11/01
WF [053000]	Woori Finance Holdings Co. Ltd	South Korea		1:3	09/03
AMX [AMXL]	America Movil	Mexico		1:20	2/01
CX [CEMEXCPO]	CEMEX	Mexico		1:10	9/99
KOF [KOFL]*	Coca Cola	Mexico		1:10	9/93
VLRS []**	Controladora ...				9/13
ICA [ICA]	Empresas	Mexico		1:4	4/92 [8/05]
FMX [FEMSAUBD]	FEMSA	Mexico		1:10	5/98 [6/05]
GMK [GRUMAB]	Gruma	Mexico		1:4	11/98 10/96]
PAC [GAPB]	GAP	Mexico		1:10	2/06 [2/06]
ASR [ASURB]	ASUR	Mexico		1:10	9/00 [9/00]
BSMX [SANMAXB]*	Grupo Financiero	Mexico		1:5	9/12 [9/12]
SIM [SIMECB]	Grupo Simec	Mexico		1:3	7/93 [02/07]
IBA [BACHOCOB]	Industrias Bachoco	Mexico		1:12	9/97
MXT [MAXCOMCP]	Maxcom Telecomunicaciones	Mexico		1:7	10/07 10/07]
CPAC [CPACASC1]*	Cementos Pacasmayo S.A.A	Peru		1:5	2/12 [7/99]
BVN [BUENAVC1]*	Compania de Minas Buenaven S.A	Peru		1:1	5/96
PHI [TELPM]	Philippines Long Distance	Philippines		1:1	10/94
MTL [MTLR]*	Mechel	Russia		1:1	10/04
MBT [MTSS]*	Mobile Telesystems	Russia		1:2	6/00 [03/09]
AU [ANG]	AngloGold Ashanti Limited	South Africa		1:1	8/98 [6/04]
DRD [DRD]	DRDGOLD Limited	South Africa		1:10	12/11
GFI [GFI]	Gold Fields Limited	South Africa		1:1	5/93 [11/93]

Ticker	Name	Primary Exchange Country	Industry	Exchange Ratio	Date of ADR [primary] listing
HMY [HAR]	Harmony Gold Mining Co Limited	South Africa		1:1	05/95 [6/95]
SPP [SAP]	Sappi Limited	South Africa		1:1	09/95 [9/95]
SSL [SOL]	Sasol Limited	South Africa		1:1	06/93
TSM [2330]	Taiwan Semiconduct. Mfg. Co Ltd	Taiwan		1:5	10/97
HIMX [3222]*	Himax Technologies Inc	Taiwan		1:2	
UMC [2303]	United Microelectronics Corp	Taiwan		1:5	09/00
AUO [2409]	AU Optronics Corp	Taiwan		1:10	05/02
SPIL [2325]	Siliconware Precision Indust. Co	Taiwan		1:5	09/00
ASX [2311]	Advanced Semiconduct. Engn. Inc	Taiwan		1:5	09/00
CHT [2412]	Chunghwa Telecom Co Ltd	Taiwan		1:10	07/03
TKC [TCELL]*	Turkcell ...A.S	Turkey		1:2.5	7/00 [7/00]

Table 3A: Descriptive statistics for ADR portfolio returns

		Midpoint	Return			Day	Return	Daily	Return	Overnight	Return
	Mean	Median	St. Dev	Kurtosis	Skew.	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev
Australia	0.01%	0.05%	1.87%	11.03	0.02	0.01%	1.35%	0.02%	2.07%	0.01%	1.43%
Belgium	0.02%	0.02%	1.92%	22.72	0.70	-0.01%	1.11%	0.02%	2.06%	0.03%	1.78%
Brazil	0.01%	0.07%	2.19%	10.95	-0.41	0.01%	1.82%	0.01%	2.45%	-0.01%	1.48%
Chile	0.02%	0.04%	1.30%	13.23	-0.41	0.01%	1.14%	0.02%	1.45%	0.01%	0.72%
Denmark	0.08%	0.08%	1.77%	15.30	-0.75	0.12%	1.08%	0.08%	1.91%	-0.05%	1.65%
Finland	0.04%	0.08%	2.92%	14.36	-0.39	0.03%	2.09%	0.04%	3.19%	0.01%	2.46%
France	-0.01%	0.03%	1.70%	9.68	-0.38	0.08%	1.22%	-0.01%	1.91%	-0.09%	1.45%
Germany	0.02%	0.00%	1.90%	17.77	0.77	0.08%	1.24%	0.02%	2.12%	-0.06%	1.67%
Hong Kong	0.00%	-0.01%	2.32%	9.39	0.39	-0.03%	1.44%	0.00%	2.61%	0.04%	1.94%
India	0.02%	0.07%	2.29%	14.18	-0.27	-0.03%	1.81%	0.02%	2.59%	0.05%	1.57%
Indonesia	0.02%	0.00%	2.87%	14.13	-0.26	0.08%	2.07%	0.02%	3.23%	-0.06%	2.48%
Ireland	0.00%	0.09%	2.72%	62.24	-3.66	-0.03%	2.07%	0.00%	3.02%	0.03%	2.16%
Israel	-0.05%	-0.06%	2.88%	29.01	-1.40	-0.19%	1.83%	-0.05%	2.92%	0.13%	2.45%
Italy	0.02%	0.03%	1.63%	20.32	0.02	0.06%	1.14%	0.02%	1.78%	-0.04%	1.38%
Japan	0.01%	0.03%	1.46%	11.37	0.23	0.08%	0.82%	0.01%	1.60%	-0.06%	1.24%
Mexico	-0.05%	0.03%	2.37%	20.96	-0.98	0.03%	2.19%	-0.05%	2.58%	-0.09%	1.40%
Netherlands	0.02%	0.08%	2.25%	10.85	-0.11	0.08%	1.49%	0.02%	2.48%	-0.05%	1.94%
Norway	0.04%	0.12%	2.06%	7.13	-0.57	-0.01%	1.41%	0.04%	2.31%	0.05%	1.75%
Philippines	0.02%	0.00%	2.00%	14.24	0.51	0.04%	1.56%	0.02%	2.21%	-0.03%	1.55%
Portugal	0.01%	0.00%	1.95%	8.03	-0.22	0.03%	1.42%	0.01%	2.21%	-0.03%	1.61%
South Africa	-0.01%	0.00%	2.03%	7.02	-0.07	-0.02%	1.72%	-0.01%	2.41%	0.01%	1.63%
South Korea	0.00%	-0.01%	2.32%	12.65	0.59	-0.01%	1.53%	0.00%	2.60%	0.01%	1.86%
Spain	0.03%	0.08%	1.85%	8.18	-0.17	0.07%	1.36%	0.03%	2.11%	-0.04%	1.56%
Switzerland	0.05%	0.03%	2.00%	8.21	0.07	0.05%	1.61%	0.05%	2.28%	-0.01%	1.65%
Taiwan	0.01%	-0.04%	2.58%	12.74	0.21	-0.01%	2.07%	0.01%	3.02%	0.03%	1.85%
UK	0.03%	0.07%	1.29%	9.67	-0.33	0.09%	0.93%	0.03%	1.44%	-0.06%	1.10%

Table 3B: Descriptive statistics, Home shares portfolios

		Midpoint	Return			Day	Return	Daily	Return	Overnight	Return
	Mean	Median	St. Dev	Kurtosis	Skew.	Mean	St. Dev	Mean	St. Dev	Mean	St. Dev
Australia	0.01%	0.03%	1.50%	10.67	-0.07	0.00%	1.17%	-1.14%	1.86%	0.01%	1.23%
Belgium	0.01%	0.03%	1.64%	23.66	0.54	-0.04%	1.72%	-1.21%	2.33%	0.04%	0.93%
Brazil	0.03%	0.06%	1.74%	12.84	-0.34	0.08%	1.66%	-1.61%	2.21%	-0.05%	1.32%
Chile	0.03%	0.03%	1.03%	9.79	-0.12	0.03%	0.90%	-1.02%	1.49%	0.00%	0.71%
Denmark	0.07%	0.07%	1.50%	11.94	-0.63	0.00%	1.66%	-1.07%	2.20%	0.07%	1.00%
Finland	0.03%	0.00%	2.74%	12.47	0.18	41.08%	75.30%	39.43%	75.48%	-41.06%	75.18%
France	-0.01%	0.05%	1.58%	10.25	-0.54	46.24%	81.08%	44.79%	81.21%	-46.26%	81.05%
Germany	0.02%	0.04%	1.70%	13.56	0.03	0.01%	1.81%	-1.35%	2.36%	0.02%	1.07%
Hong Kong	0.00%	0.01%	2.35%	10.69	0.06	-0.11%	2.13%	-2.02%	3.04%	0.11%	1.47%
India	0.04%	0.07%	1.96%	12.82	-0.73	-0.07%	1.90%	-1.95%	2.59%	0.11%	1.11%
Indonesia	0.06%	0.00%	2.53%	10.28	0.19	0.01%	2.46%	-1.44%	3.29%	0.05%	1.51%
Ireland	0.00%	0.09%	2.71%	55.02	-3.40	-1.40%	4.36%	-3.15%	4.91%	1.40%	4.13%
Israel	-0.06%	-0.08%	2.46%	22.59	-1.00	-0.08%	1.81%	-1.45%	2.94%	0.02%	2.02%
Italy	0.02%	0.06%	1.37%	14.78	-0.67	128.1%	283.9%	126.8%	283.8%	-128.1%	283.8%
Japan	0.01%	0.01%	1.52%	7.76	-0.08	-0.06%	1.11%	-1.28%	1.86%	0.07%	1.25%
Mexico	-0.02%	0.04%	1.99%	20.18	-0.81	0.06%	1.80%	-1.42%	2.57%	-0.08%	1.30%
Netherlands	0.02%	0.10%	2.06%	9.51	-0.42	19.61%	34.30%	18.20%	34.59%	-19.60%	34.21%
Norway	0.02%	0.07%	1.64%	5.93	-0.56	-0.05%	1.59%	-1.07%	2.25%	0.07%	1.12%
Philippines	0.03%	0.00%	1.93%	13.25	0.52	-0.01%	1.62%	-0.88%	2.43%	0.04%	1.40%
Portugal	0.01%	0.01%	1.79%	12.97	-0.13	99.60%	207.3%	98.40%	207.3%	-99.61%	207.2%
South Africa	0.02%	0.00%	1.81%	6.58	0.17	0.91%	4.55%	-0.58%	4.82%	-0.88%	4.51%
South Korea	0.02%	0.02%	1.85%	8.06	0.07	-0.01%	1.70%	-1.64%	2.43%	0.02%	1.46%
Spain	0.03%	0.07%	1.39%	8.46	-0.28	101.1%	175.5%	99.8%	175.6%	-101.1%	175.5%
Switzerland	0.04%	0.07%	1.86%	8.60	-0.17	27.79%	142.5%	26.21%	141.8%	-27.76%	142.5%
Taiwan	0.01%	0.01%	1.96%	7.78	-0.25	-0.09%	1.69%	-1.43%	2.45%	0.10%	1.63%
UK	0.03%	0.07%	1.15%	9.17	-0.48	-0.02%	1.10%	-1.23%	1.60%	0.05%	0.77%

Table 3C: First Order Auto Correlation Coefficients for Midpoint, Day, Daily, and Overnight Returns

	ADR				Home Shares			
	Midpoint	Day	Daily	Overnight	Midpoint	Day	Daily	Overnight
Australia	0.118	-0.023	-0.023	0.003	0.067	-0.025	-0.085	-0.045
Belgium	0.074	0.032	0.027	0.050	0.214	0.110	-0.024	-0.004
Brazil	0.189	0.000	-0.009	-0.013	0.186	0.065	0.051	0.066
Chile	0.271	0.129	0.114	-0.010	0.255	0.210	0.082	-0.010
Denmark	0.055	0.064	0.050	0.082	0.211	0.045	-0.063	0.006
Finland	0.109	-0.014	-0.014	-0.001	0.165	0.998	-0.003	0.999
France	0.128	-0.008	-0.007	0.031	0.167	0.999	-0.018	0.999
Germany	0.099	0.029	0.038	0.081	0.190	-0.003	-0.072	0.023
Hong Kong	0.149	-0.053	-0.045	0.087	0.219	0.016	-0.069	-0.002
India	0.160	0.055	0.063	-0.025	0.127	-0.019	-0.068	0.014
Indonesia	0.129	0.002	0.002	0.090	0.166	-0.009	-0.054	0.019
Ireland	0.161	0.023	0.023	0.042	0.142	0.632	0.025	0.697
Israel	0.045	0.069	0.063	0.060	0.142	0.098	-0.059	0.044
Italy	0.040	0.031	0.029	0.015	0.169	0.999	-0.063	0.999
Japan	0.111	-0.034	-0.030	0.041	0.140	-0.051	-0.126	-0.014
Mexico	0.217	0.068	0.070	-0.017	0.205	0.083	0.062	0.011
Netherlands	0.111	-0.047	-0.046	0.002	0.182	0.996	-0.027	0.998
Norway	0.056	-0.050	-0.042	-0.004	0.125	-0.066	-0.169	-0.047
Philippines	0.194	-0.013	-0.013	0.111	0.162	0.062	-0.049	0.034
Portugal	0.119	0.007	0.015	0.067	0.157	0.999	-0.018	0.999
South Africa	0.178	-0.029	-0.028	0.072	0.146	0.385	-0.006	0.405
South Korea	0.093	-0.003	0.006	-0.015	0.174	-0.056	-0.110	-0.007
Spain	0.142	-0.055	-0.054	0.018	0.170	0.990	-0.030	0.990
Switzerland	0.149	-0.009	-0.009	0.028	0.199	0.996	-0.015	0.996
Taiwan	0.152	0.026	0.035	-0.023	0.238	-0.147	-0.132	0.161
UK	0.110	-0.001	-0.002	-0.012	0.139	0.013	-0.108	-0.109

Table 3D: Joint Test of Equality of Mean and Variance

The OLS estimates reported in the table below pertain to the parameters in the following regression equations:

$$r_{Q,t}^A - r_{Q,t}^H = a + b(r_{Q,t}^A + r_{Q,t}^H - [r_{Q,t}^A + r_{Q,t}^H]) + e_{Q,t}^A$$

Table 4: Descriptive statistics for Volatility and Spread

	Volatility						Spread					
	Mean-ADR	Median-ADR	St. dev-ADR	Mean-Home	Median-Home	St. dev-Home	Mean-ADR	Median-ADR	St. dev-ADR	Mean-Home	Median-Home	St. dev-Home
Australia	0.05%	0.01%	0.21%	0.04%	0.02%	0.08%	0.27	0.25	0.06	0.29	0.28	0.05
Belgium	0.01%	0.01%	0.03%	0.03%	0.02%	0.10%	0.25	0.24	0.05	0.30	0.29	0.06
Brazil	0.06%	0.04%	0.09%	0.07%	0.04%	0.08%	0.34	0.33	0.06	0.35	0.34	0.05
Chile	0.03%	0.02%	0.05%	0.03%	0.02%	0.05%	0.28	0.27	0.05	0.26	0.26	0.06
Denmark	0.01%	0.01%	0.02%	0.03%	0.01%	0.07%	0.25	0.24	0.05	0.30	0.29	0.06
Finland	0.05%	0.02%	0.08%	0.06%	0.03%	0.14%	0.33	0.32	0.07	0.34	0.33	0.08
France	0.02%	0.01%	0.04%	0.05%	0.03%	0.06%	0.28	0.27	0.05	0.32	0.32	0.05
Germany	0.02%	0.01%	0.03%	0.04%	0.02%	0.11%	0.26	0.25	0.06	0.32	0.31	0.07
Hong Kong	0.03%	0.02%	0.06%	0.08%	0.05%	0.14%	0.29	0.28	0.06	0.36	0.35	0.06
India	0.05%	0.02%	0.10%	0.09%	0.05%	0.14%	0.32	0.31	0.06	0.36	0.35	0.06
Indonesia	0.04%	0.02%	0.10%	0.05%	0.02%	0.12%	0.31	0.30	0.07	0.33	0.32	0.07
Ireland	0.08%	0.03%	0.28%	0.12%	0.04%	0.58%	0.32	0.32	0.07	0.33	0.34	0.09
Israel	0.04%	0.01%	0.18%	0.04%	0.02%	0.07%	0.28	0.28	0.10	0.32	0.31	0.07
Italy	0.02%	0.01%	0.14%	0.03%	0.02%	0.04%	0.27	0.26	0.05	0.31	0.30	0.05
Japan	0.02%	0.01%	0.04%	0.03%	0.02%	0.04%	0.25	0.25	0.05	0.31	0.31	0.04
Mexico	0.09%	0.05%	0.18%	0.06%	0.03%	0.18%	0.34	0.33	0.07	0.31	0.31	0.07
Netherlands	0.03%	0.01%	0.06%	0.05%	0.02%	0.11%	0.28	0.27	0.06	0.32	0.31	0.07
Norway	0.02%	0.01%	0.04%	0.02%	0.01%	0.04%	0.29	0.28	0.05	0.29	0.28	0.06
Philippines	0.02%	0.01%	0.05%	0.02%	0.01%	0.04%	0.27	0.27	0.06	0.26	0.26	0.07
Portugal	0.02%	0.01%	0.04%	0.03%	0.01%	0.06%	0.28	0.27	0.07	0.29	0.29	0.07
South Africa	0.05%	0.03%	0.08%	0.05%	0.03%	0.06%	0.32	0.32	0.07	0.33	0.33	0.06
South Korea	0.04%	0.02%	0.07%	0.06%	0.03%	0.07%	0.29	0.28	0.06	0.34	0.33	0.05
Spain	0.02%	0.01%	0.04%	0.03%	0.02%	0.04%	0.28	0.27	0.06	0.31	0.30	0.05
Switzerland	0.03%	0.01%	0.07%	0.05%	0.03%	0.08%	0.29	0.28	0.06	0.32	0.31	0.06
Taiwan	0.05%	0.03%	0.10%	0.04%	0.02%	0.04%	0.33	0.33	0.06	0.32	0.31	0.05
UK	0.02%	0.01%	0.03%	0.04%	0.02%	0.05%	0.26	0.25	0.05	0.31	0.30	0.05

Table 5A: VAR estimates of the parameters of ADR midpoint return

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_{5_t}$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_{5_t}$	OVR	OVR_t	R ²
Australia	0.164	14.50*	-0.019	-2.85*	-0.073	-5.42*	0.000	0.00	1.050	96.84*	0.66
Belgium	0.123	7.77*	0.000	0.00	-0.124	-6.75*	0.000	0.00	0.940	101.93*	0.77
Brazil	0.153	8.08*	-0.028	-3.28*	0.050	2.10^	0.000	0.00	1.130	75.12*	0.61
Chile	0.213	12.05*	0.000	0.00	0.000	0.00	0.000	0.00	1.224	69.35*	0.54
Denmark	0.080	6.79*	0.000	0.00	-0.053	-3.89*	-0.022	-2.65*	0.916	117.47*	0.74
Finland	0.191	12.98*	0.038	3.21*	-0.170	-11.56*	-0.028	-2.39^	0.947	95.55*	0.67
France	0.239	16.72*	0.000	0.00	-0.200	-13.42*	0.000	0.00	0.949	102.59*	0.69
Germany	0.099	7.43*	0.017	1.78#	-0.033	-2.27^	-0.038	-3.96*	0.948	101.54*	0.71
Hong Kong	0.168	14.71*	0.000	0.00	-0.112	-10.76*	-0.024	-2.59^	1.045	129.07*	0.78
India	0.214	15.22*	-0.017	-2.10^	-0.095	-5.99*	0.000	0.00	1.148	75.77*	0.65
Indonesia	0.146	11.74*	-0.047	-6.72*	-0.082	-5.97*	0.034	4.27*	0.938	92.72*	0.68
Ireland	0.236	15.18*	0.000	0.00	-0.162	-10.84*	0.000	0.00	0.978	89.36*	0.65
Israel	0.000	0.00	-0.046	-3.57*	0.123	6.15*	0.037	2.76*	0.901	60.53*	0.63
Italy	0.085	5.53*	0.000	0.00	-0.052	-2.87^	0.000	0.00	0.918	82.07*	0.60
Japan	0.122	13.13*	0.022	3.77*	-0.088	-10.28*	-0.014	-2.65*	1.056	144.27*	0.82
Mexico	0.000	0.00	0.000	0.00	0.258	10.27*	0.000	0.00	1.052	58.30*	0.46
Netherlands	0.196	14.28*	0.035	3.53*	-0.164	-11.41*	-0.040	-3.90*	0.979	117.06*	0.74
Norway	0.169	10.14*	0.000	0.00	-0.168	-8.24*	0.000	0.00	0.971	82.76*	0.70
Philippines	0.143	10.58*	0.000	0.00	-0.069	-5.39*	-0.021	-2.13^	0.996	86.16*	0.64
Portugal	0.162	10.72*	0.027	2.53^	-0.105	-6.54*	-0.032	-2.89*	0.954	88.36*	0.64
South Africa	0.221	16.36*	0.014	1.98^	-0.153	-10.35*	0.000	0.00	0.920	80.63*	0.59
South Korea	0.153	15.00*	-0.020	-3.27*	-0.118	-9.49*	0.012	1.71#	1.058	112.60*	0.74
Spain	0.205	16.67*	0.015	2.27^	-0.174	-10.79*	-0.021	-2.51^	0.967	104.20*	0.70
Switzerland	0.251	14.62*	0.000	0.00	-0.156	-8.74*	-0.027	-2.36^	0.913	74.85*	0.61
Taiwan	0.241	18.21*	-0.014	-1.68#	-0.124	-7.29*	0.000	0.00	1.106	81.81*	0.65
UK	0.239	17.10*	-0.020	-2.41^	-0.190	-12.29*	0.000	0.00	0.946	100.91*	0.68

*, ^, and # indicate less than 1%, 5%, and 10% levels of significance.

Table 5A: VAR estimates of the parameters of ADR daily return

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_5_t$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_5_t$	OVR	OVR_t	R ²
Australia	-0.013	-1.12	-0.024	-3.74*	0.026	1.93#	0.013	2.26^	1.097	81.45*	0.57
Belgium	0.029	1.78#	-0.025	-3.16*	-0.036	-2.29^	0.023	4.20*	0.968	85.55*	0.70
Brazil	-0.020	-1.07	-0.003	-0.30	0.022	1.03	-0.008	-0.96	1.110	55.07*	0.45
Chile	0.086	5.16*	-0.009	-1.24	-0.008	-0.44	0.010	1.83#	1.236	54.83*	0.40
Denmark	-0.010	-0.81	-0.008	-1.23	0.002	0.17	-0.004	-0.95	0.951	101.31*	0.68
Finland	0.000	0.01	-0.004	-0.74	-0.013	-1.57	0.003	1.64	0.976	78.56*	0.57
France	-0.005	-0.49	-0.006	-1.21	-0.019	-3.54*	0.004	3.66*	1.003	83.87*	0.59
Germany	-0.028	-2.00#	0.001	0.17	0.030	2.19^	-0.019	-3.95*	1.020	88.88*	0.65
Hong Kong	-0.016	-1.35	-0.022	-3.41*	-0.002	-0.19	0.005	1.07	1.122	104.34*	0.70
India	-0.002	-0.11	-0.017	-2.28^	0.031	1.91#	0.008	1.36	1.216	62.24*	0.54
Indonesia	-0.036	-2.86*	-0.030	-4.40*	0.032	2.54^	0.004	0.67	0.996	77.86*	0.59
Ireland	-0.036	-2.65*	0.013	2.38^	0.005	0.46	0.002	0.73	1.009	71.24*	0.52
Israel	0.007	0.40	-0.011	-1.04	0.083	4.36*	-0.004	-0.50	0.895	56.39*	0.60
Italy	-0.016	-1.47	0.001	0.23	-0.003	-2.03^	0.001	2.09^	0.981	78.73*	0.58
Japan	-0.016	-1.60	-0.002	-0.29	-0.010	-1.16	0.004	1.01	1.106	115.70*	0.74
Mexico	-0.064	-3.12*	0.017	1.73#	0.118	5.45*	-0.005	-0.62	0.944	41.65*	0.28
Netherlands	-0.014	-1.15	-0.012	-2.73*	-0.007	-0.71	0.002	0.79	1.017	93.21*	0.64
Norway	-0.032	-2.02^	-0.038	-3.89*	0.003	0.17	0.018	2.41^	1.036	71.43*	0.63
Philippines	-0.043	-3.21*	0.004	0.43	0.019	1.55	-0.016	-2.58*	1.010	67.35*	0.50
Portugal	-0.005	-0.44	-0.007	-1.38	-0.005	-2.00#	0.001	2.02^	1.043	79.38*	0.58
South Africa	-0.009	-0.75	-0.002	-0.38	-0.007	-0.94	0.001	0.36	1.032	68.10*	0.49
South Korea	-0.005	-0.41	-0.028	-4.64*	-0.018	-1.49	0.012	2.43^	1.124	92.74*	0.66
Spain	-0.030	-2.92*	-0.002	-0.37	-0.002	-1.38	0.000	1.25	1.032	83.04*	0.58
Switzerland	0.019	1.50	-0.026	-4.39*	-0.003	-1.73#	0.001	1.83	0.974	61.95*	0.50
Taiwan	0.021	1.50	-0.019	-2.22^	0.013	0.73	0.001	0.19	1.217	67.32*	0.54
UK	-0.006	-0.48	-0.012	-1.72#	-0.020	-1.53	-0.002	-0.38	0.988	82.22*	0.58

Table 5B: VAR estimates of the parameters of Home midpoint return

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_{5_t}$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_{5_t}$	OVR	OVR_t	R ²
Australia	-0.035	-3.09*	-0.019	-2.71*	0.162	16.70*	0.023	4.16*	0.864	77.20*	0.66
Belgium	-0.181	-6.88*	0.000	0.00	0.398	17.53*	0.026	1.74#	0.804	31.53*	0.35
Brazil	0.123	5.36*	0.000	0.00	0.073	4.00*	0.000	0.00	0.824	50.38*	0.43
Chile	0.081	4.57*	0.000	0.00	0.156	11.10*	0.014	2.09^	0.952	67.03*	0.54
Denmark	-0.118	-6.70*	0.000	0.00	0.422	27.71*	0.000	0.00	0.670	40.07*	0.39
Finland	-0.411	-21.15*	-0.145	-9.23*	0.671	34.53*	0.160	10.10*	0.000	0.00	0.34
France	-0.366	-16.72*	-0.042	-3.03*	0.633	30.30*	0.041	3.05*	0.000	0.00	0.23
Germany	-0.213	-11.20*	-0.044	-3.44*	0.489	27.82*	0.042	3.34*	0.590	30.48*	0.38
Hong Kong	-0.130	-8.91*	-0.040	-3.11*	0.467	28.54*	0.027	1.88#	0.862	53.50*	0.60
India	-0.073	-4.26*	-0.073	-7.09*	0.237	14.77*	0.067	7.60*	0.877	35.62*	0.45
Indonesia	-0.068	-4.30*	0.000	0.00	0.336	22.61*	0.000	0.00	0.829	41.07*	0.47
Ireland	-0.378	-18.39*	-0.085	-5.54*	0.654	30.69*	0.093	6.04*	0.164	20.76*	0.33
Israel	0.000	0.00	-0.070	-5.56*	0.117	7.34*	0.058	4.85*	0.845	49.94*	0.57
Italy	-0.175	-7.93*	-0.032	-2.37^	0.417	22.05*	0.000	0.00	0.000	0.00	0.15
Japan	-0.052	-4.87*	0.000	0.00	0.312	26.38*	0.000	0.00	0.901	94.86*	0.75
Mexico	0.168	8.09*	-0.032	-2.90*	0.000	0.00	0.026	2.80*	0.990	61.68*	0.47
Netherlands	-0.395	-17.86*	-0.099	-6.25*	0.651	30.70*	0.101	6.49*	0.000	0.00	0.27
Norway	-0.180	-9.33*	0.000	0.00	0.363	23.06*	0.000	0.00	0.908	52.03*	0.58
Philippines	-0.078	-6.03*	-0.022	-2.32^	0.358	24.76*	0.019	2.07^	0.788	53.36*	0.64
Portugal	-0.308	-14.38*	-0.064	-4.33*	0.574	28.34*	0.059	4.20*	0.000	-1.91#	0.23
South Africa	-0.177	-9.79*	-0.020	-2.09^	0.462	27.92*	0.000	0.00	0.067	13.19*	0.23
South Korea	-0.031	-2.49^	0.000	0.00	0.290	27.09*	-0.015	-2.42^	0.755	58.95*	0.59
Spain	-0.167	-8.45*	-0.054	-5.16*	0.388	25.76*	0.020	2.57^	0.000	-2.65*	0.19
Switzerland	-0.342	-15.37*	-0.056	-3.89*	0.664	31.02*	0.060	4.11*	0.000	0.00	0.30
Taiwan	0.000	0.00	-0.022	-2.89*	0.156	14.16*	0.028	4.28*	0.799	58.45*	0.65
UK	-0.160	-9.35*	0.000	0.00	0.415	26.86*	0.000	0.00	0.887	59.48*	0.53

*, ^, and # indicate less than 1%, 5%, and 10% levels of significance.

Table 5B: VAR estimates of the parameters of Home daily return

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_{5_t}$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_{5_t}$	OVR	OVR_t	R ²
Australia	0.115	8.06*	0.078	13.43*	0.129	9.48*	-0.048	-7.41*	0.860	45.19*	0.45
Belgium	0.239	8.61*	0.093	9.62*	0.143	4.91*	-0.087	-6.23*	0.797	20.45*	0.26
Brazil	0.299	13.53*	0.027	3.17*	0.045	2.30^	-0.024	-2.52^	0.749	32.42*	0.31
Chile	0.341	19.84*	0.072	13.70*	0.152	9.65*	-0.055	-7.98*	0.988	45.67*	0.50
Denmark	0.271	13.45*	0.050	6.45*	0.199	9.00*	-0.079	-6.69*	0.710	25.69*	0.26
Finland	0.321	13.12*	0.061	15.65*	0.530	18.71*	-0.129	-11.04*	-0.373	-13.54*	1.00
France	0.634	20.22*	0.028	8.79*	0.126	3.42*	-0.111	-7.87*	-0.228	-7.29*	1.00
Germany	0.265	11.66*	0.079	9.93*	0.157	6.56*	-0.110	-9.42*	0.663	21.69*	0.25
Hong Kong	0.185	10.78*	0.112	15.68*	0.255	12.14*	-0.187	-17.53*	0.971	35.79*	0.40
India	0.247	12.34*	0.041	5.90*	0.133	6.66*	-0.008	-0.86	0.736	19.47*	0.31
Indonesia	0.166	9.58*	0.049	6.71*	0.217	11.51*	-0.067	-7.34*	0.825	25.45*	0.30
Ireland	-0.035	-1.90#	0.167	30.20*	0.744	32.99*	-0.197	-19.26*	-0.091	-4.66*	0.53
Israel	0.254	11.39*	0.023	2.25^	0.061	2.77*	-0.044	-3.40*	0.852	37.20*	0.45
Italy	0.593	4.96*	0.001	0.31	0.164	1.21	-0.100	-1.86#	-0.401	-3.35*	1.00
Japan	0.169	12.13*	0.078	13.11*	0.175	10.27*	-0.111	-13.00*	0.969	58.67*	0.56
Mexico	0.225	10.93*	0.038	4.79*	0.061	3.12*	-0.021	-2.20^	0.922	39.57*	0.35
Netherlands	0.390	16.37*	0.137	29.46*	0.303	11.61*	-0.219	-21.80*	0.074	2.92*	0.99
Norway	0.180	8.19*	0.070	7.44*	0.177	8.33*	-0.080	-6.518	0.997	32.48*	0.38
Philippines	0.121	8.18*	0.035	4.71*	0.299	15.59*	-0.042	-4.42*	0.784	30.69*	0.43
Portugal	0.608	9.38*	0.004	1.50	0.175	2.31^	-0.122	-4.12*	-0.371	-5.72*	1.00
South Africa	0.006	0.56	0.025	7.17*	0.627	34.59*	-0.019	-2.31^	-0.821	-80.56*	0.70
South Korea	0.205	13.24*	0.053	8.41*	0.176	11.61*	-0.088	-11.51*	0.815	37.37*	0.38
Spain	-0.001	-0.09	0.001	0.76	0.522	11.58*	-0.018	-0.84	-0.995	-215.6*	1.00
Switzerland	0.756	7.14*	0.002	0.77	-0.035	-0.29	-0.021	-0.45	-0.225	-2.13*	0.99
Taiwan	0.174	10.95*	0.023	3.28*	0.033	2.38^	-0.013	-1.67	0.884	39.03*	0.45
UK	0.212	11.68*	0.091	14.76*	0.231	12.04*	-0.095	-10.14*	0.889	35.81*	0.39

Table 6A: VAR estimates of the parameters of ADR Volatility

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_5_t$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_5_t$	AVol	AVol_t	R ²
Australia	0.201	14.85*	0.027	5.62*	0.000	0.00	0.000	0.00	1.748	49.11*	0.46
Belgium	0.126	8.58*	0.039	8.25*	0.007	1.95#	0.000	0.00	0.567	61.02*	0.65
Brazil	0.184	10.88*	0.078	15.85*	0.052	2.96*	0.000	0.00	0.566	46.71*	0.68
Chile	0.127	8.32*	0.049	10.06*	0.143	10.10*	0.025	5.16*	0.610	49.82*	0.66
Denmark	0.066	5.35*	0.051	12.57*	0.018	4.73*	0.000	0.00	0.538	66.61*	0.59
Finland	0.119	10.67*	0.045	12.02*	0.012	2.27^	0.004	2.03^	0.471	89.39*	0.72
France	0.158	12.09*	0.028	6.40*	0.022	2.28^	0.027	9.32*	0.582	72.94*	0.74
Germany	0.050	3.49*	0.047	8.85*	0.000	0.00	0.000	0.00	0.530	49.84*	0.46
Hong Kong	0.115	11.77*	0.073	24.43*	0.052	11.32*	-0.005	-3.52*	0.427	101.02*	0.78
India	0.128	9.81*	0.056	13.96*	0.015	1.81#	0.014	5.32*	0.480	61.91*	0.69
Indonesia	0.000	0.00	0.045	10.54*	0.039	4.28*	0.000	0.00	0.530	89.51*	0.74
Ireland	0.117	13.36*	0.044	13.13*	0.000	0.00	0.004	2.38^	0.780	126.58*	0.81
Israel	0.000	0.00	0.015	1.68#	0.000	0.00	0.000	0.00	0.603	19.70*	0.15
Italy	0.000	0.00	0.000	0.00	0.000	0.00	0.075	3.24*	0.533	9.05*	0.04
Japan	0.067	3.94*	0.073	13.24*	0.130	6.48*	0.000	0.00	0.466	21.47*	0.37
Mexico	0.104	6.79*	0.062	12.46*	0.070	4.98*	0.000	0.00	0.526	43.66*	0.45
Netherlands	0.141	12.30*	0.034	8.12*	0.020	2.89*	0.019	7.64*	0.519	94.87*	0.82
Norway	0.074	4.76*	0.063	12.78*	0.111	8.26*	0.015	2.96*	0.410	52.65*	0.76
Philippines	0.047	4.64*	0.031	8.61*	0.000	0.00	0.000	0.00	0.503	93.83*	0.69
Portugal	0.067	6.04*	0.071	18.81*	0.000	0.00	0.000	0.00	0.453	86.80*	0.71
South Africa	0.080	6.53*	0.077	19.38*	0.000	0.00	0.021	3.96*	0.522	72.52*	0.71
South Korea	0.214	16.50*	0.038	9.96*	0.059	4.41*	0.000	0.00	0.594	63.96*	0.70
Spain	0.136	12.44*	0.048	13.20*	0.057	4.27*	0.017	4.23*	0.442	88.99*	0.76
Switzerland	0.181	11.98*	0.018	3.02*	0.044	3.59*	0.023	5.35*	0.464	55.43*	0.62
Taiwan	0.112	10.47*	0.054	15.87*	0.100	3.57*	0.036	4.07*	0.384	89.61*	0.76
UK	0.188	13.44*	0.041	8.91*	0.028	3.08*	0.014	4.88*	0.563	59.53*	0.70

*, ^, and # indicate less than 1%, 5%, and 10% levels of significance.

Table 6B: VAR estimates of the parameters of Home Volatility

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_5_t$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_5_t$	HVol	HVol_t	R ²
Australia	0.065	3.95*	0.079	14.02*	0.057	9.94*	0.005	2.60*	0.088	11.42*	0.28
Belgium	0.065	5.47*	0.019	4.13*	0.000	0.00	0.000	0.00	0.561	81.30*	0.71
Brazil	0.115	5.80*	0.069	10.25*	0.105	5.53*	0.000	0.00	0.155	18.97*	0.41
Chile	0.238	12.82*	0.063	9.98*	0.233	11.74*	-0.021	-3.37*	0.145	12.08*	0.42
Denmark	0.067	4.81*	0.032	5.90*	0.000	0.00	0.000	0.00	0.449	46.34*	0.37
Finland	0.068	4.78*	0.021	3.60*	0.178	5.97*	0.030	3.04*	0.248	50.54*	0.42
France	0.182	10.53*	0.119	23.37*	0.215	9.28*	-0.050	-6.65*	0.165	29.93*	0.63
Germany	0.048	3.06*	0.044	7.66*	0.113	2.24^	0.070	3.80*	0.417	36.80*	0.34
Hong Kong	0.200	12.47*	0.071	13.86*	-0.061	-1.76#	0.000	0.00	0.266	26.97*	0.37
India	0.248	12.64*	0.052	8.24*	0.341	11.17*	-0.016	-1.72#	0.125	11.88*	0.35
Indonesia	0.247	16.52*	0.016	2.95*	0.170	9.06*	0.000	0.00	0.150	41.84*	0.51
Ireland	0.073	6.33*	0.000	0.00	0.219	8.88*	0.068	7.29*	0.525	80.17*	0.62
Israel	0.000	0.00	0.099	12.47*	0.000	0.00	0.014	3.52*	0.021	5.46*	0.13
Italy	0.275	16.95*	0.073	17.26*	0.009	2.79*	0.000	0.00	0.142	29.60*	0.53
Japan	0.372	22.35*	0.058	12.19*	-0.045	-3.16*	0.008	1.84#	0.149	20.77*	0.50
Mexico	0.086	5.16*	0.016	2.40^	0.087	4.75*	0.000	0.00	0.280	21.13*	0.16
Netherlands	0.188	11.37*	0.053	8.80*	0.317	11.23*	0.000	0.00	0.203	40.12*	0.63
Norway	0.134	7.04*	0.043	6.19*	0.083	3.80*	0.020	2.98*	0.126	28.54*	0.52
Philippines	0.276	16.62*	0.034	6.18*	0.000	0.00	0.000	0.00	0.057	14.50*	0.19
Portugal	0.166	9.88*	0.025	3.95*	0.095	4.20*	0.028	3.66*	0.124	20.93*	0.23
South Africa	0.244	14.77*	0.071	13.99*	0.119	9.97*	0.000	0.00	0.064	16.46*	0.45
South Korea	0.245	14.23*	0.082	17.21*	0.114	6.81*	0.000	0.00	0.078	16.85*	0.50
Spain	0.194	11.93*	0.091	18.66*	0.134	10.11*	-0.015	-3.38*	0.094	27.89*	0.56
Switzerland	0.230	12.53*	0.062	9.60*	0.241	10.55*	0.000	0.00	0.167	19.35*	0.46
Taiwan	0.172	9.16*	0.080	13.77*	0.026	3.69*	0.005	2.39^	0.024	10.88*	0.31
UK	0.138	7.95*	0.094	17.34*	0.278	10.22*	0.000	0.00	0.200	20.12*	0.55

*, ^, and # indicate less than 1%, 5%, and 10% levels of significance.

Table 6C: VAR estimates of the parameters of Home Spread

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_5_t$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_5_t$	AVol	AVol_t	R ²
Australia	0.400	25.52*	0.078	18.99*	0.000	0.00	0.018	3.66*	5.913	27.29*	0.77
Belgium	0.336	19.14*	0.044	9.43*	0.000	0.00	0.011	2.71*	84.269	43.89*	0.77
Brazil	0.425	24.33*	0.051	11.52*	-0.050	-2.46^	-0.023	-4.14*	25.115	37.61*	0.69
Chile	0.411	25.28*	0.060	15.80*	0.000	0.00	0.000	0.00	32.840	37.23*	0.74
Denmark	0.348	25.15*	0.032	8.90*	0.000	0.00	0.013	4.23*	96.800	48.33*	0.68
Finland	0.278	20.90*	0.041	10.07*	0.000	0.00	0.020	5.77*	45.578	66.36*	0.80
France	0.398	27.58*	0.025	6.32*	0.000	0.00	0.030	8.03*	49.108	51.41*	0.83
Germany	0.340	23.59*	0.045	11.27*	0.023	1.96#	0.014	4.30*	73.176	49.70*	0.75
Hong Kong	0.411	28.09*	0.060	17.02*	0.000	0.00	0.000	0.00	31.521	42.98*	0.79
India	0.321	20.87*	0.043	10.06*	0.000	0.00	0.016	3.69*	28.803	49.82*	0.78
Indonesia	0.367	25.06*	0.055	14.18*	0.000	0.00	0.000	0.00	31.448	47.12*	0.73
Ireland	0.350	20.56*	0.074	15.95*	0.027	1.91#	0.011	3.15*	6.495	30.73*	0.68
Israel	0.370	17.85*	0.069	12.79*	0.000	0.00	0.000	0.00	20.524	27.29*	0.57
Italy	0.438	26.75*	0.052	10.96*	0.070	3.89*	0.017	3.49*	11.436	36.15*	0.70
Japan	0.462	29.42*	0.070	18.62*	0.042	2.53^	0.000	0.00	25.359	27.37*	0.82
Mexico	0.211	12.82*	0.061	12.95*	0.103	6.29*	0.000	0.00	15.626	46.17*	0.64
Netherlands	0.395	26.48*	0.025	5.71*	0.025	1.80#	0.037	9.47*	31.461	45.33*	0.84
Norway	0.374	21.02*	0.026	5.28*	0.042	2.59*	0.020	4.16*	51.654	35.06*	0.76
Philippines	0.375	26.82*	0.047	13.17*	0.000	0.00	0.000	0.00	53.299	47.40*	0.65
Portugal	0.342	24.14*	0.049	13.48*	0.000	0.00	0.013	3.99*	61.556	53.42*	0.79
South Africa	0.254	17.37*	0.043	9.77*	-0.038	-2.08^	0.052	10.64*	32.028	42.10*	0.67
South Korea	0.397	27.17*	0.059	16.78*	0.000	0.00	0.009	2.40^	28.815	42.77*	0.82
Spain	0.338	24.03*	0.046	12.10*	0.026	1.80#	0.027	6.74*	47.572	51.65*	0.83
Switzerland	0.311	19.24*	0.053	11.40*	0.000	0.00	0.034	7.04*	34.371	49.78*	0.86
Taiwan	0.290	19.69*	0.059	15.49*	0.036	2.37^	0.017	4.12*	26.898	50.16*	0.78
UK	0.433	28.41*	0.028	6.99*	0.000	0.00	0.034	9.51*	43.003	43.31*	0.86

*, ^, and # indicate less than 1%, 5%, and 10% levels of significance.

Table 6D: VAR estimates of the parameters of Home Spread

	α_1	α_{1_t}	$\Sigma\alpha_5$	$\Sigma\alpha_5_t$	β_1	β_{1_t}	$\Sigma\beta_5$	$\Sigma\beta_5_t$	HVol	HVol_t	R ²
Australia	0.373	26.70*	0.038	9.47*	0.093	7.47*	0.000	0.00	21.128	40.15*	0.72
Belgium	0.399	21.62*	0.036	6.81*	0.110	4.91*	0.011	1.79#	22.365	34.79*	0.69
Brazil	0.211	14.88*	0.034	8.78*	0.040	3.29*	0.000	0.00	36.272	65.20*	0.76
Chile	0.435	26.70*	0.067	17.70*	0.111	6.09*	-0.028	-6.61*	29.627	31.53*	0.72
Denmark	0.409	30.22*	0.053	15.13*	0.072	4.66*	0.000	0.00	36.262	48.18*	0.72
Finland	0.206	17.27*	0.037	8.96*	0.151	9.94*	0.020	4.31*	30.114	76.71*	0.78
France	0.291	20.45*	0.068	18.44*	0.094	6.73*	-0.014	-3.63*	36.377	53.88*	0.86
Germany	0.341	23.51*	0.048	11.70*	0.115	6.70*	0.000	0.00	26.069	48.72*	0.73
Hong Kong	0.330	24.95*	0.049	14.43*	0.121	8.44*	-0.012	-3.37*	19.304	57.09*	0.77
India	0.229	15.73*	0.054	13.38*	0.083	5.62*	0.000	0.00	20.816	56.38*	0.80
Indonesia	0.341	24.03*	0.033	8.20*	0.081	5.33*	0.000	0.00	27.071	47.16*	0.68
Ireland	0.322	18.26*	0.106	23.91*	0.124	5.96*	-0.020	-3.53*	4.192	32.55*	0.73
Israel	0.307	17.00*	0.045	9.55*	0.000	0.00	-0.006	-1.80#	48.112	42.66*	0.68
Italy	0.276	20.24*	0.049	13.35*	0.044	3.63*	-0.009	-2.63*	71.961	61.18*	0.82
Japan	0.326	24.53*	0.063	20.06*	0.090	7.42*	-0.012	-4.04*	47.238	53.21*	0.85
Mexico	0.361	20.68*	0.072	14.87*	0.000	0.00	0.000	0.00	12.444	35.19*	0.60
Netherlands	0.335	22.06*	0.074	17.27*	0.100	6.11*	-0.015	-3.29*	20.648	44.54*	0.84
Norway	0.318	19.32*	0.028	5.92*	0.035	1.97^	0.014	2.88*	65.667	44.63*	0.76
Philippines	0.375	27.06*	0.031	8.69*	0.109	7.36*	-0.009	-2.35^	70.526	44.92*	0.63
Portugal	0.361	26.18*	0.053	14.78*	0.065	4.16*	0.000	0.00	52.246	52.82*	0.78
South Africa	0.227	16.10*	0.079	21.38*	0.058	5.22*	0.000	0.00	43.050	49.34*	0.79
South Korea	0.251	19.69*	0.053	16.33*	0.031	2.66*	0.000	0.00	37.434	64.05*	0.82
Spain	0.293	22.63*	0.049	14.49*	0.043	3.60*	0.000	0.00	61.674	61.86*	0.86
Switzerland	0.282	17.05*	0.060	12.47*	0.104	6.37*	0.009	1.93#	24.623	43.72*	0.84
Taiwan	0.204	17.57*	0.042	14.15*	-0.020	-1.85#	0.014	4.97*	84.302	78.54*	0.84
UK	0.337	22.53*	0.090	24.02*	0.148	9.48*	-0.037	-9.26*	30.638	45.70*	0.88

*, ^, and # indicate less than 1%, 5%, and 10% levels of significance.

Table 8: Granger Bi-directional and Instantaneous Causality Tests for ADR and Home Shares Returns, Volatility, and Spread

	H0: Home equity returns do not Granger-cause ADR returns	H0: ADR returns do not Granger-cause Home equity returns	H0: No Instant. causality between ADR and Home equity returns	H0: Home equity variance do not Granger-cause ADR variance	H0: ADR variance do not Granger-cause Home equity variance	H0: No Instant. causality between Home equity and ADR variance	H0: Home equity spread do not Granger-cause ADR spread	H0: ADR spread do not Granger-cause Home equity spread	H0: No Instant. causality between: Home equity and ADR spread
	F-stat	F-stat	Chi-Square	F-stat	F-stat	Chi-Square	F-stat	F-stat	Chi-Square
Australia	1.22	312.76*	1166.98*	17.65*	68.67*	118.41*	11.17*	62.47*	116.39*
Belgium	2.00	192.26*	1291.69*	2.82*	19.92*	192.75*	5.97*	26.44*	433.56*
Brazil	4.29*	4.02*	1464.99*	9.08*	26.81*	546.74*	3.55*	16.64*	526.88*
Chile	4.98*	51.08*	1777.06*	54.22*	35.42*	790.28*	3.47*	17.29*	714.37*
Denmark	2.43^	427.24*	1819.53*	41.83*	16.42*	111.80*	12.39*	14.73*	296.20*
Finland	24.71*	516.67*	1895.09*	4.41*	45.73*	579.33*	8.87*	79.62*	673.54*
France	18.76*	327.01*	1940.68*	42.58*	23.56*	822.58*	15.61*	37.92*	769.07*
Germany	2.50^	282.51*	1583.03*	11.70*	21.16*	141.07*	11.12*	36.46*	317.35*
Hong Kong	5.08*	494.75*	1848.30*	26.54*	14.63*	176.70*	3.96*	44.52*	256.68*
India	3.51*	197.81*	774.70*	8.65*	45.02*	127.17*	10.69*	44.74*	94.92*
Indonesia	5.24*	319.52*	1196.17*	37.00*	56.57*	479.97*	4.61*	38.42*	196.66*
Ireland	10.85*	336.90*	1776.27*	2.42^	42.83*	619.76*	9.90*	20.56*	532.58*
Israel	19.51*	50.95*	679.96*	1.66	4.13*	4.67	1.30	2.36^	25.33*
Italy	0.89	207.27*	1665.36*	17.12*	4.07*	23.79*	19.12*	25.19*	579.90*
Japan	2.05^	317.22*	1486.66*	33.29*	7.89*	329.98*	8.47*	43.58*	160.87*
Mexico	27.28*	7.69*	1945.90*	11.18*	25.03*	504.93*	24.14*	3.98*	1004.43*
Netherlands	15.93*	329.33*	2064.34*	60.21*	43.61*	1026.54*	22.55*	37.21*	776.50*
Norway	6.78*	139.68*	1162.48*	52.73*	39.31*	128.74*	20.80*	25.14*	228.51*
Philippines	2.58*	599.31*	1340.68*	6.51*	5.93*	99.83*	2.49^	22.04*	137.03*
Portugal	2.83*	308.14*	1727.13*	9.13*	26.55*	392.32*	8.67*	25.47*	491.23*
South Africa	13.46*	222.16*	1383.10*	19.78*	56.78*	304.33*	34.79*	29.34*	254.15*
South Korea	6.60*	298.28*	1122.61*	14.80*	43.72*	151.76*	8.65*	28.74*	120.31*
Spain	14.02*	182.56*	1646.70*	28.67*	29.64*	492.29*	17.95*	34.05*	587.19*
Switzerland	10.27*	290.32*	1460.34*	17.00*	89.75*	450.49*	16.57*	38.63*	601.82*
Taiwan	6.63*	313.75*	1010.21*	20.60*	21.57*	26.11*	7.06*	26.26*	42.60*
UK	12.20*	195.37*	1882.01*	20.45*	73.45*	600.00*	15.69*	45.15*	732.76*

*, ^, and # indicate less than 1%, 5%, and 10% levels of significance.

Table 9: Portfolio Returns and Differences-in-mean Returns

		Portfolio return		Difference in portfolio returns		
	Mean	Median	St. Dev		t_statistic	p_value
Large-cap portfolio	0.04%	0.08%	1.30%	Mean: Large-Small	1.530	0.126
Mid-cap portfolio	0.04%	0.06%	1.70%	Mean: Large-Mid	-0.398	0.690
Small-cap portfolio	0.01%	0.07%	1.53%	Mean: Mid-Small	1.877	0.061
	Large-Small		Large-Mid		Mid-Small	
	t_statistic	p_value	t_statistic	p_value	t_statistic	p_value
1993-94	0.964	0.336	-0.402	0.688	1.556	0.120
1995-96	3.327	0.001	-0.068	0.946	2.659	0.008
1997-98	0.133	0.894	-0.811	0.418	0.926	0.355
1999-00	0.636	0.525	0.040	0.968	0.702	0.483
2001-02	-0.687	0.492	0.603	0.547	-1.089	0.277
2003-04	-0.272	0.786	0.038	0.969	-0.662	0.508
2005-06	-0.879	0.380	0.282	0.778	-1.026	0.305
2007-08	0.024	0.981	-0.840	0.401	0.742	0.458
2009-10	0.224	0.823	0.481	0.631	-0.470	0.638
2011-12	0.255	0.799	-0.218	0.828	0.440	0.660

Table 10: VAR for Large, Mid, and Small-cap Portfolio Returns

	α_1	α_{1_t}	$\Sigma\alpha_5$	β_1	β_{1_t}	$\Sigma\beta_5$	Granger causality	F-test	p-value
Large	-0.046	-2.345	-0.122	0.000	0.000	0.003	H0: small do not Granger-cause large	3.924	0.001
Small	-0.052	-2.276	-0.096	0.088	0.020	0.093	H0: large do not Granger-cause small	2.402	0.035
Large	0.000	0.000	0.000	-0.030	0.015	-0.030	H0: mid do not Granger-cause large	1.567	0.166
Mid	0.000	0.000	0.000	0.000	0.000	-0.040	H0: large do not Granger-cause mid	0.363	0.874
Mid	-0.041	-2.050	-0.092	0.039	0.022	0.012	H0: small do not Granger-cause mid	3.863	0.002
Small	0.000	0.000	0.000	0.077	0.020	0.085	H0: mid do not Granger-cause small	1.088	0.365